9.6 Strange Dice

WORKSHEET

The question

STEP I 1993, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s1-q1

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself If the first die has p_i faces showing i and the second has q_j faces showing j, how many ways are there to throw a total of t?

STAGE 1 The extremes

Ask yourself There is no way to throw 2, nor 12. What does that say about the 1s and the 6s? There *are* two ways to throw 3. Which die has the 1s?

STAGE 2 Working upwards

Ask yourself Write down the equation for a total of 4, remembering $b_1 = 0$. What does it say in each of the two cases?

STAGE 3 Case (a) fails

Ask yourself In case (a), write down the equation for a total of 7, and use $b_5 + b_6 = 5$.

STAGE 4 Case (b) succeeds

Ask yourself In case (b), use the totals 6 and 11, and the fact that $a_6 b_6 = 0$.

Try this next

1. **Tetrahedral dice.** (From STEP III, 2003.) Standard four-faced dice are labelled $1, 2, 3, 4$. Find all the ways in which two four-faced dice can be labelled with positive integers so that every total has the same probability as for two standard four-faced dice. (*Hint:* $x + x^2 + x^3 + x^4 = x(1 + x)(1 + x^2)$.)

2. **Checking the table.** Multiply out $(x + x^4 + x^5 + 3x^6)(2x^2 + x^3 + x^4 + 2x^5)$ and check it against the table in the question.

3. **Three dice.** Are there three six-sided dice, not all standard, labelled with positive integers, whose totals have the same distribution as three standard dice? (*Hint:* the factors of $(x + \cdots + x^6)^3$ now include three copies of $1 - x + x^2$.)

4. **A coin and an octahedron.** An eight-sided die labelled $1, \ldots, 8$ and a two-sided "die" (a coin) labelled $1, 2$: can they be relabelled differently with the same totals?

Further reading

- **Martin Gardner, *Penrose Tiles to Trapdoor Ciphers*** (Mathematical Association of America). Includes the column on Sicherman dice.
- **Joseph Gallian & David Rusin, "Cyclotomic polynomials and nonstandard dice"**, *Discrete Mathematics* 27 (1979). The general theory, for dice with any number of faces.
- **Herbert Wilf, *generatingfunctionology*** (free online), Chapter 1.

The full guided solution, and what the question leads to at university, is in Section 9.6:
https://beyondstep.keithmcnulty.org/ch09/strange-dice.html